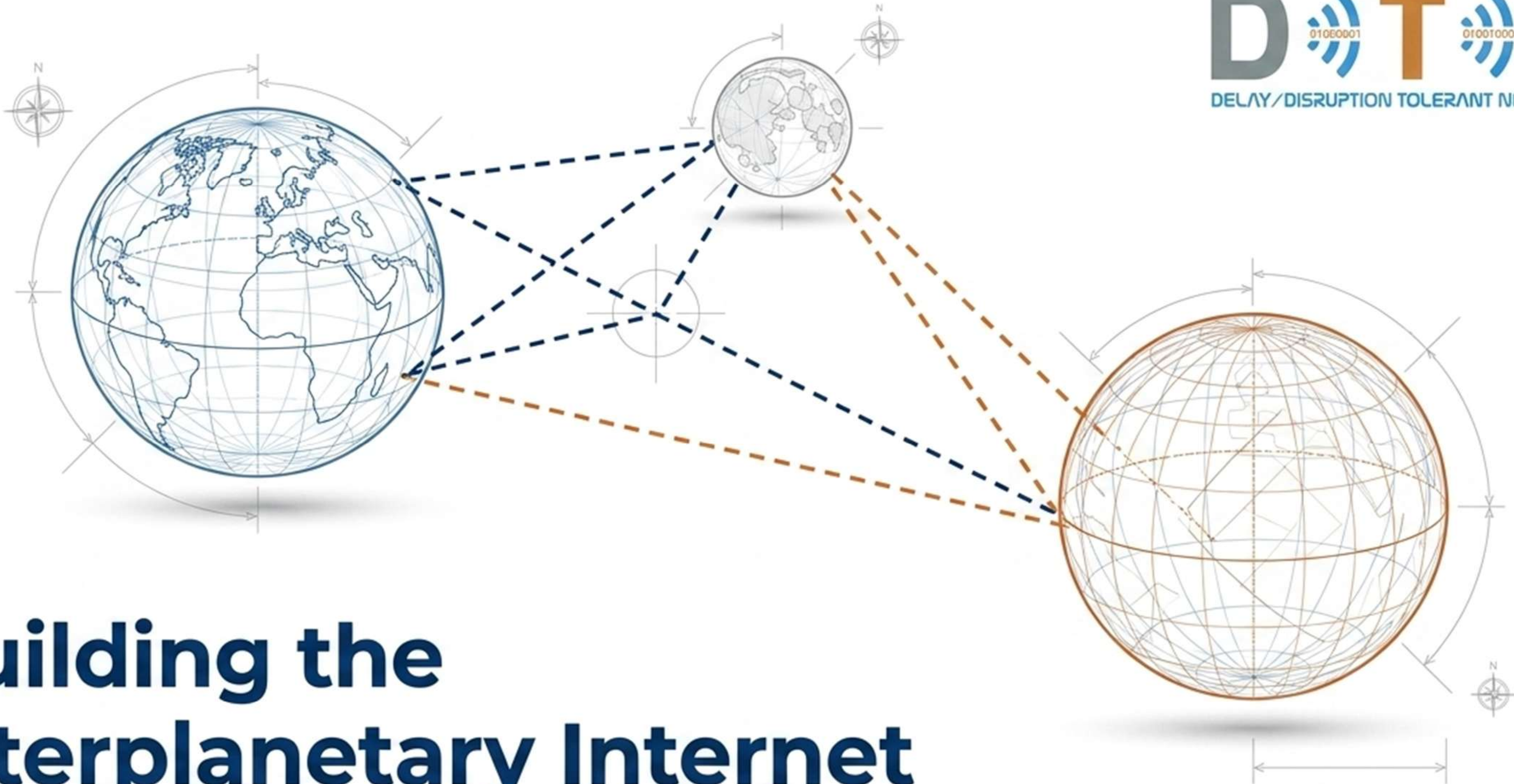
DTN

DELAY/DISRUPTION TOLERANT NETWORKING



Building the Interplanetary Internet

A Blueprint for Delay/Disruption Tolerant Networking (DTN)

The Solar System Demands a Paradigm Shift

Social

Sustained human presence on the Moon and Mars requires reliable life-support telemetry and personal communication links.

Political

International space agency cooperation relies on shared, standardized infrastructure to interoperate securely.

Technical

Autonomous robotics and advanced sensor arrays demand robust, automated data return without manual link management.

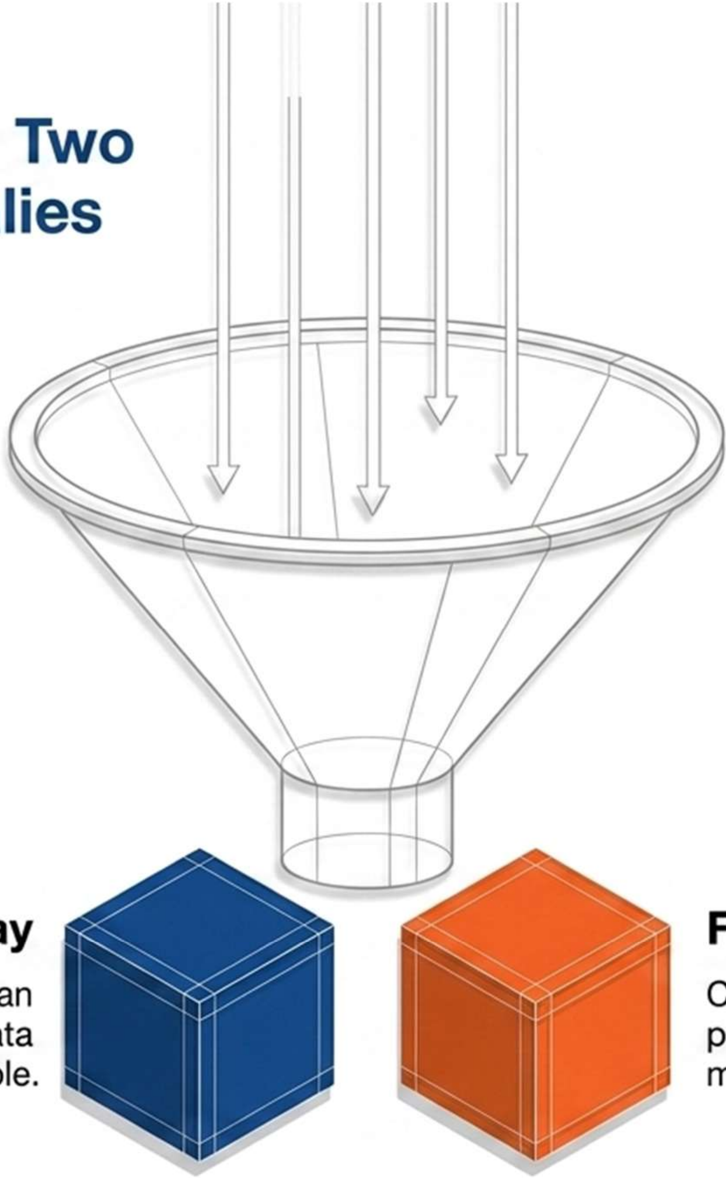
Economic

Commercial spaceflight and resource extraction depend on commoditized, scalable network architectures to reduce risk and cost.

Five Physical Realities of Space Communication

- Extreme Distances (up to 44.6 minute Round Trip Time to Mars)
- Constant Disruption (orbital mechanics cause frequent LOS/AOS cycles)
- Asymmetric Data Rates (6 Mbps down to Earth vs. 1 Kbps up to spacecraft)
- Weak Signals (spacecraft operate on limited power with small antennae)
- High Error Rates (vulnerability to cosmic radiation and solar interference)

Distilling Physics into Two Core Network Anomalies



Extreme Delay

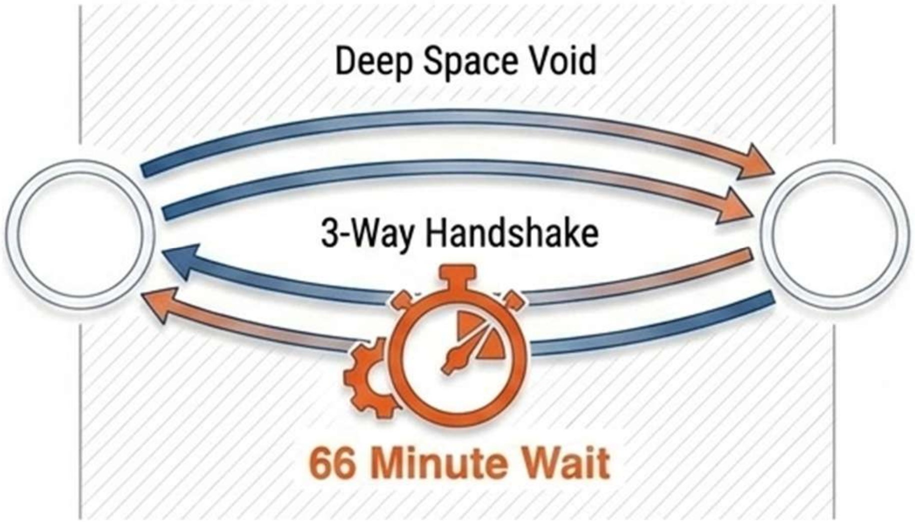
Speed-of-light limitations mean interactive, synchronous data requests are physically impossible.

Frequent Disruption

Continuous end-to-end network paths rarely exist at any given moment due to orbital occlusions.

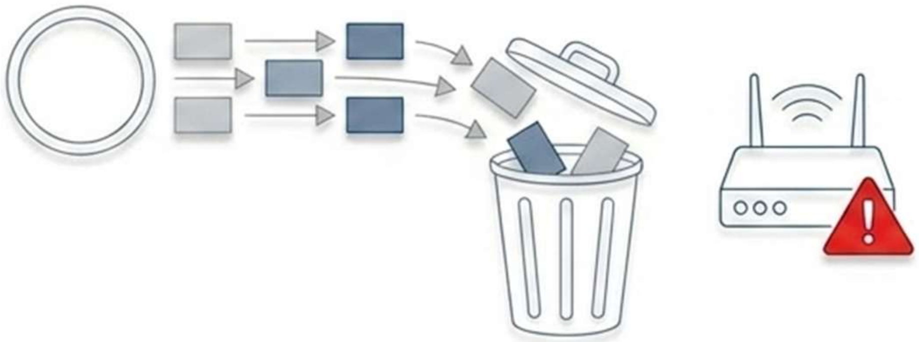
Traditional Internet Protocols Cannot Survive the Void

TCP Failure



TCP Freezes: TCP requires a chatty 3-way handshake to connect. Over Mars distances, this consumes 1.5 RTTs (up to 66 minutes) before a single byte of data is sent. Any lost packet freezes the entire stream.

UDP & Routing Failure



UDP Drops: UDP lacks flow control. It sends data blindly, resulting in massive data loss when orbital links go down.

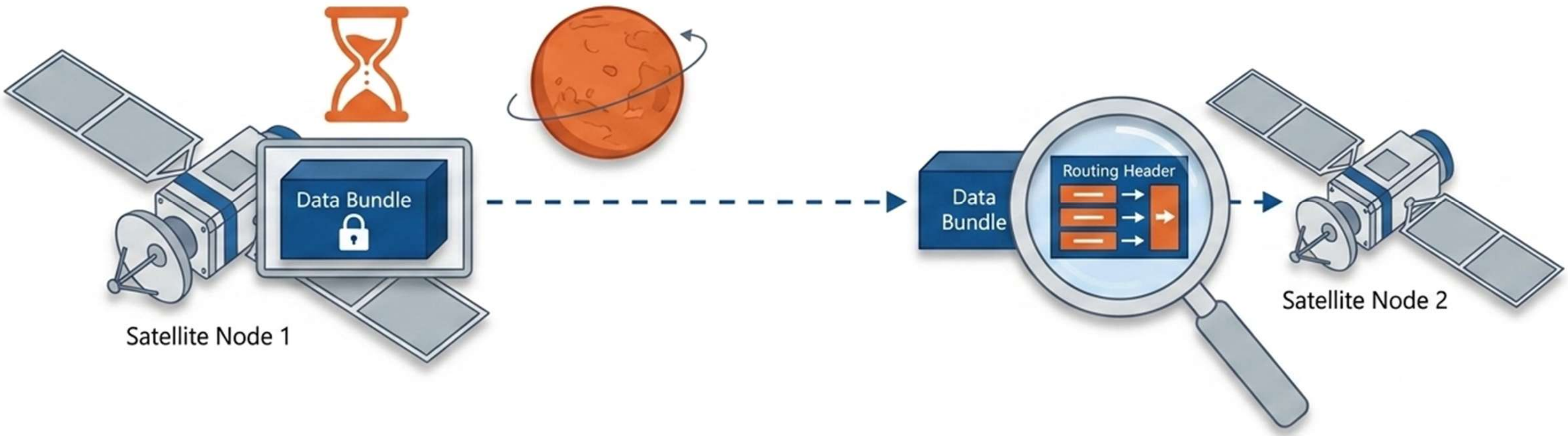
Routing Fails: BGP assumes continuous paths. It misinterprets transient orbital blockages as permanent topology failures.

NotebookLM

DTN Acts as a Smart Post Office for Space

Solving Disruption (Store-and-Forward)

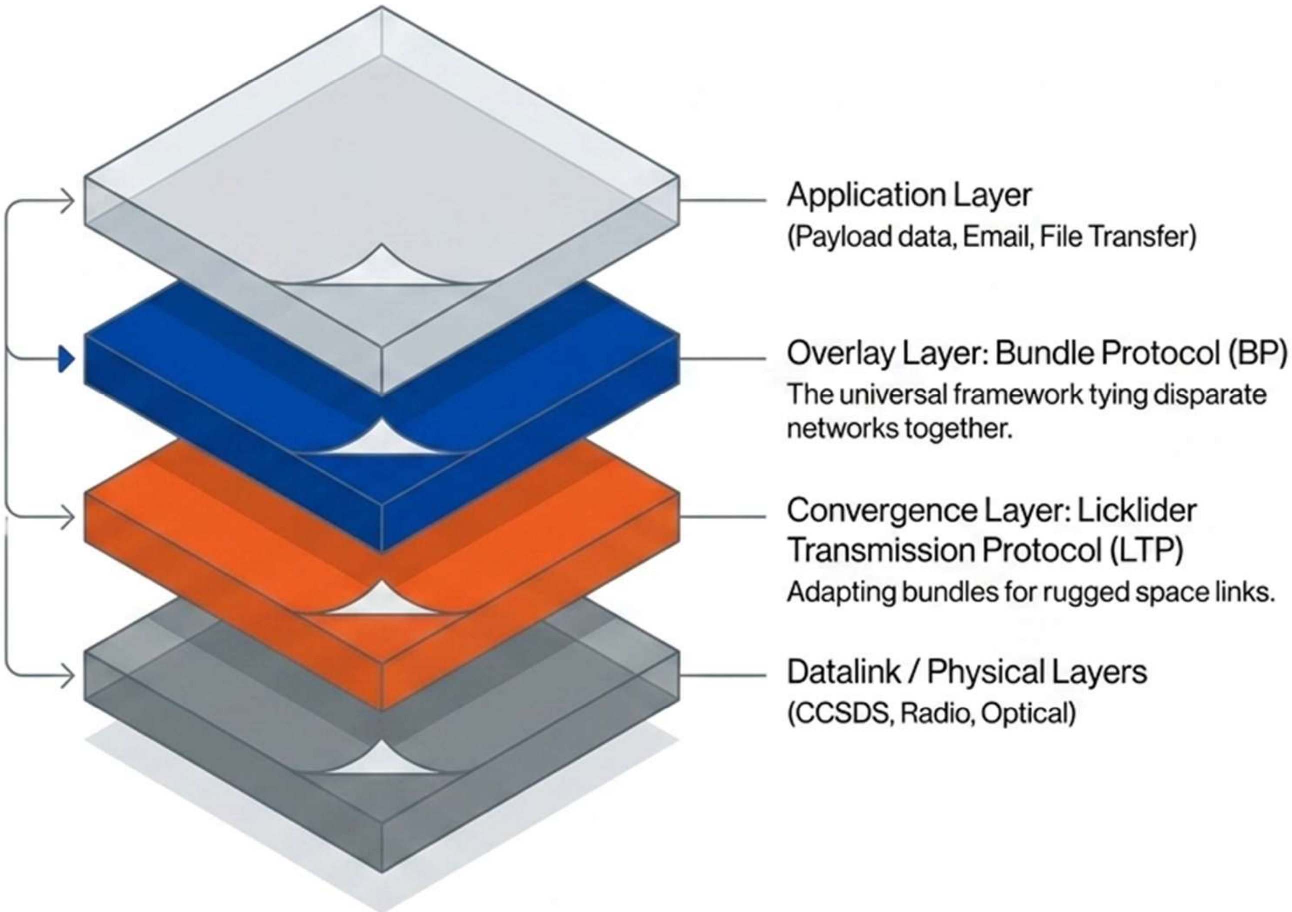
Solving Delay (Overlay Bundling)



Nodes safely store data locally on disk indefinitely until the next orbital link becomes available, progressing incrementally without needing an end-to-end path.

Data is grouped into distinct bundles containing all necessary destination routing headers. This eliminates chatty connection setups, allowing data to move forward autonomously.

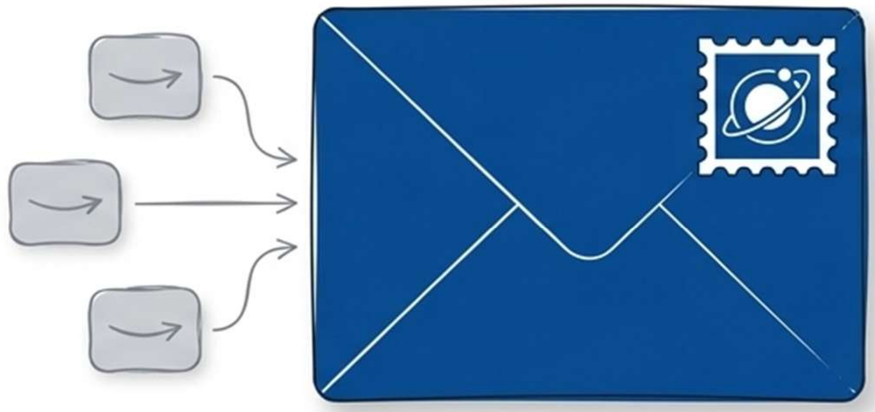
The Interplanetary Protocol Stack



NotebookLM

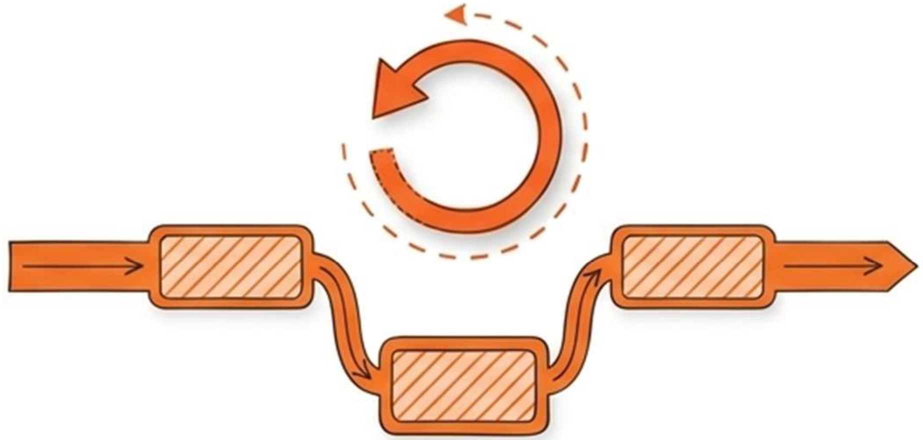
The Engines of DTN: BP and LTP

BP Bundle



Bundle Protocol (BP): The overlay layer. It aggregates data into self-contained bundles, handling the intelligent routing and executing the store-and-forward mechanism across multiple planetary hops.

LTP Segmentation



Licklider Transmission Protocol (LTP): The convergence layer designed specifically for deep-space. It divides BP blocks into segments, automatically detects transmission failures, and concurrently retransmits only the lost segments over heavily disrupted links.

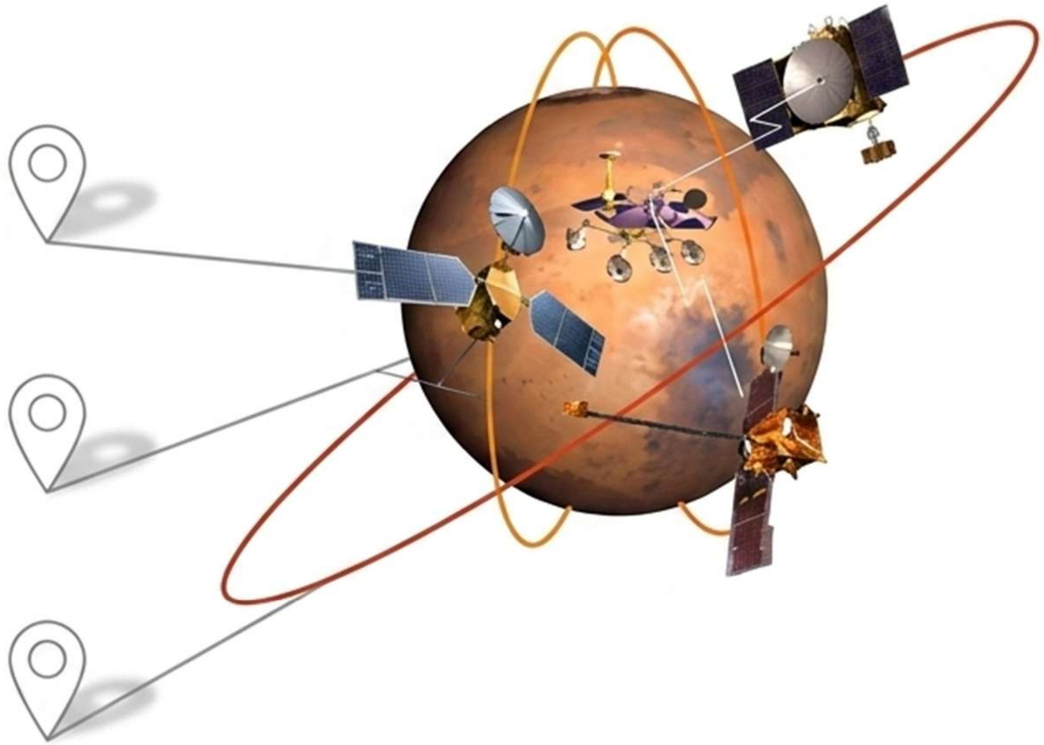
Interplanetary Overlay Network (ION) Deployment

ION is NASA's open-source C/C++ software distribution of the DTN architecture.

International Space Station
Reliably delivers critical health, safety, and scientific payload data.

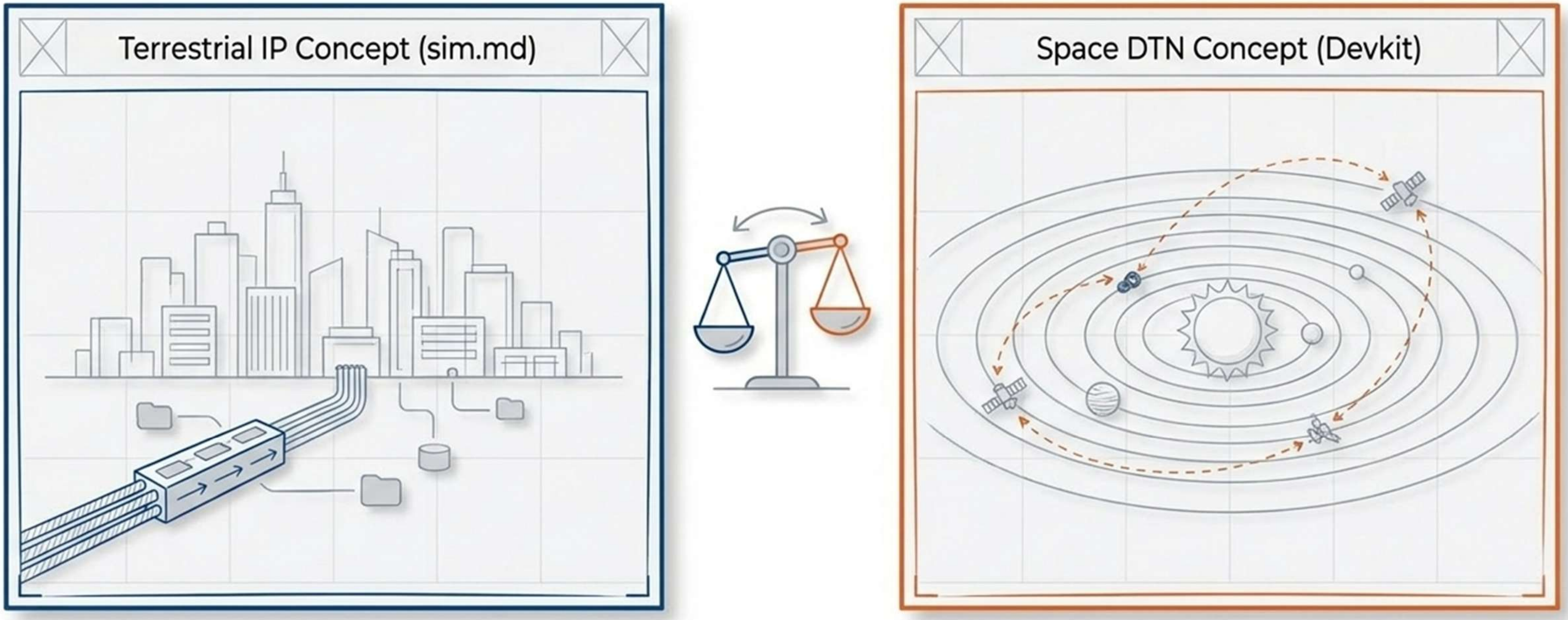
SmallSat/CubeSat Constellations
Coordinates and automates data delivery from distributed, low-power spacecraft in Low Earth Orbit.

Deep Space Missions
Maximizes link efficiency for distant orbiters over extreme time delays.

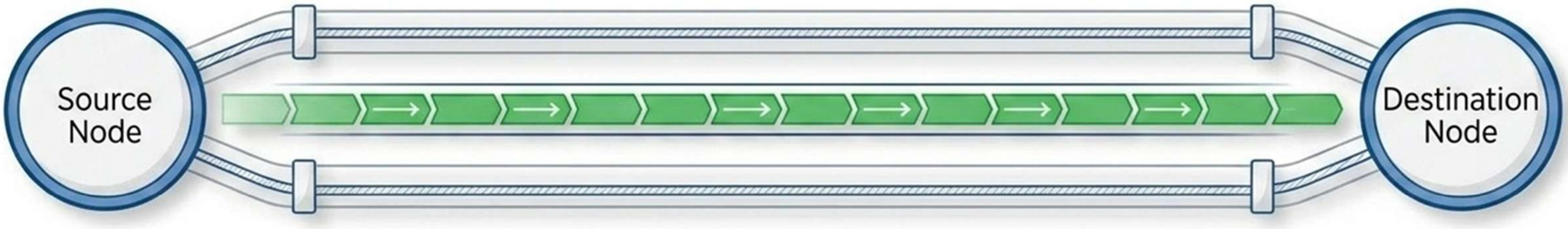


Simulating the Two Network Paradigms

Testing the theory requires comparing standard terrestrial synchronous networks against deep-space asynchronous realities.



Earth Simulation: Continuous Synchronous Flow



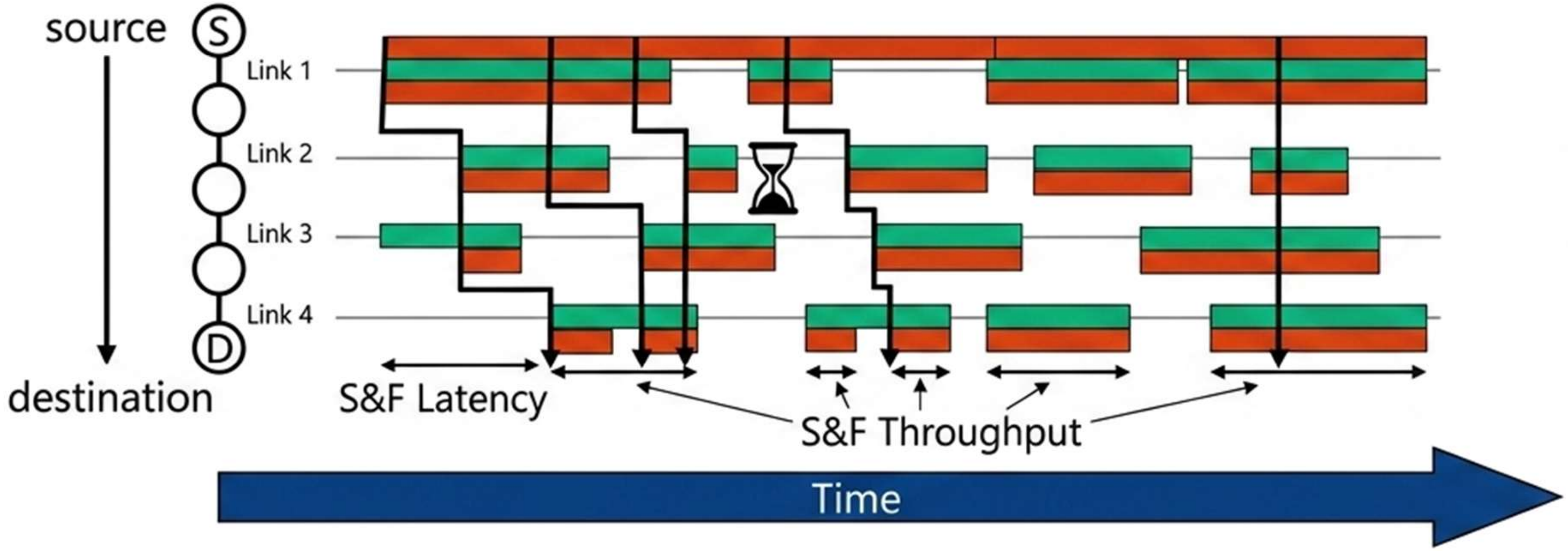
Environment (sim.md)

Short round-trip times (100–300ms) and continuous, unbroken connections.

Outcome

Standard TCP/UDP tools instantly establish a full end-to-end path. High throughput is achieved immediately because acknowledgments return in milliseconds and the routing topology never changes.

EarthMars Simulation: Patient, Incremental Progress



Environment (DTN Devkit)

Extreme latency (44 minute RTT), asymmetric data rates, and frequent planetary obstruction.

Outcome

IP models fail entirely. ION uses LTP to queue bundles locally. The node patiently waits for scheduled Line-of-Sight (LOS), then safely forwards the aggregated data segments. Data reaches Mars successfully without ever requiring an end-to-end handshake.

The Foundation for the Solar System Internet

1. The Need
Expanding human and robotic presence across the solar system.

2. The Limit
TCP/IP shatters under extreme delay and constant disruption.

3. The Architecture
DTN introduces Store-and-Forward overlay bundling.

4. The Protocols
BP handles the routing; LTP guarantees the rugged links.

5. The Reality
ION safely and reliably connects missions from the ISS to Deep Space today.

Building the Solar System Internet

Questions & Discussion